StockSense AI

Sales forecasts for clearer stock decisions

One fictional retailer. Five anonymous products.

Seven forecast days. Web and terminal access.

VUT AI Solutions

The retailer’s stock problem

Current process: check recent sales and choose an order manually.

Too little stock can lead to shortages and missed sales.

Too much stock ties money up in slow-selling products.

Intended community benefit: more reliable access to goods.

The manager needs an estimate before placing an order

Objectives and success criteria

Forecast: seven dated estimates for each of five products.

Accuracy: lower daily error than the weekly benchmark.

Stock planning: show the calculation and reject invalid inputs.

Text assistant: recognise at least 20 of 25 supported requests.

Success means a working system with measured evidence

Requirements, constraints and risks

Required: product selection, history, forecasts and stock inputs.

Scope: one store, five products and a seven-day horizon.

Risks: future data entering training and invalid stock inputs.

Responses: time-based splits, validation and saved local models.

A small scope makes the solution easier to test and explain

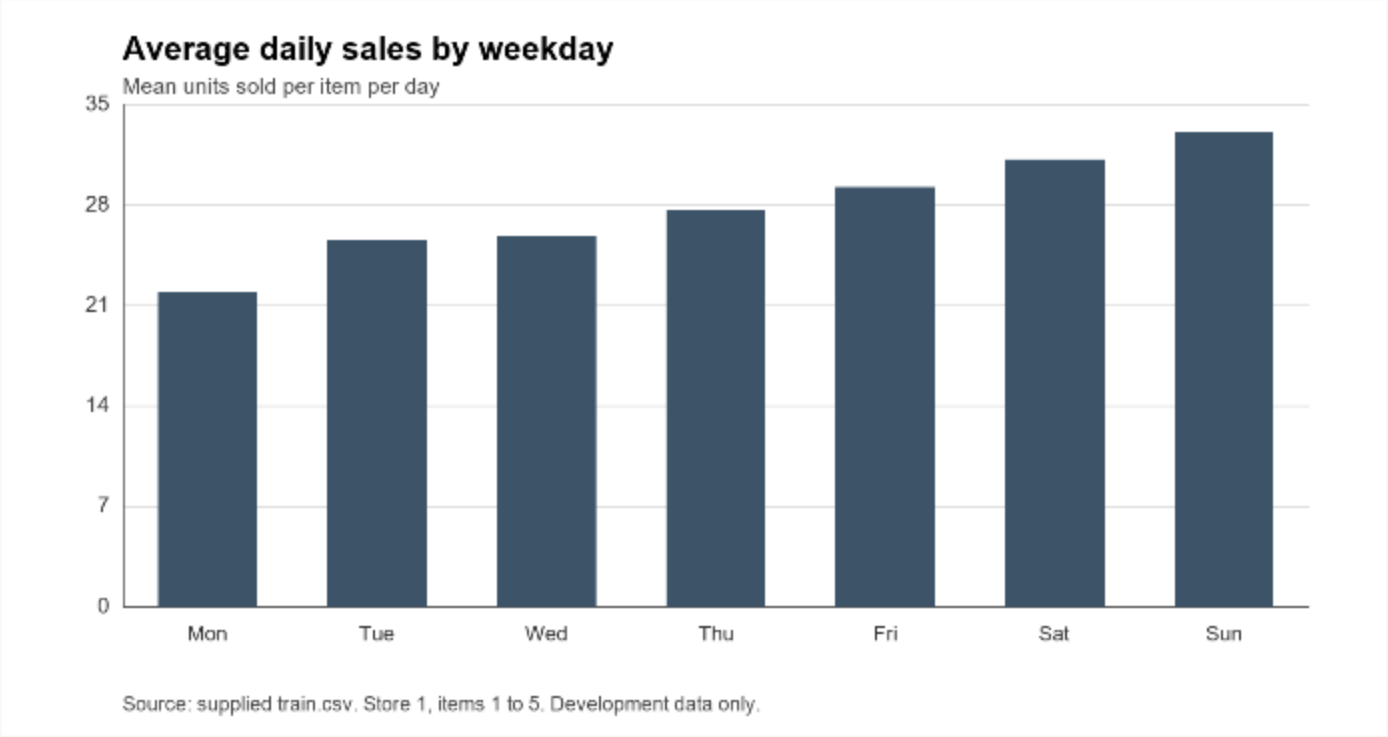
The dataset and its four fields

2013–2017 **913,000 source rows** **9,130 selected rows**

Field	Type	Example	Meaning
date	Temporal	2013-01-01	Day of the sale
store	Categorical ID	1	Anonymous store
item	Categorical ID	1	Anonymous product
sales	Numerical count	13	Units sold that day

Store and product numbers identify categories, not quantities.

The data shows a weekly pattern



Monday

21.88 units

Sunday

33.01 units

Weekday patterns help us choose useful forecast inputs.

How past sales become a forecast

1. Select store 1 and products 1 to 5.

2. Check dates and sales, then sort each product’s history.

3. Use past sales, recent averages and calendar information.

4. Predict the next seven daily sales quantities.

Features are the inputs. Future daily sales are the targets.

Training, validation and final testing

2013–2015

TRAIN

Learn from history

2016

VALIDATE

Choose the method

2017

TEST

Check later results

We chose the model before looking at the final test results

The three forecasting methods

Weekly benchmark	Repeat last week’s sales on matching weekdays.
Random forest	Combine decision trees using sales and calendar features.
Small LSTM	Use a neural network to learn from a sequence of sales.

The benchmark tells us whether machine learning adds value

The LSTM deep-learning comparison

28

past sales days

16

LSTM units

7

predicted days

Early stopping checks whether validation error still improves.

The LSTM learns sequence patterns from past daily sales

How we chose the model

MAE = average error size

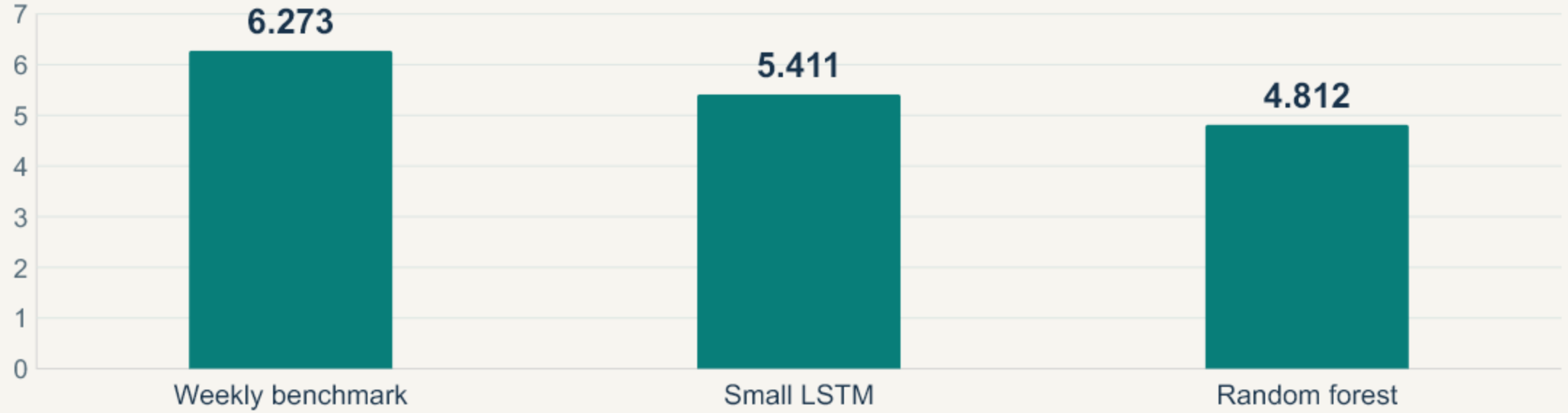
Predict 12 units
Observe 10 units
Absolute error = 2 units



2016 daily MAE in units. Lower is better.

Random forest had the lowest daily validation MAE: 4.911 units

The final test confirmed lower daily error



Daily MAE in units. Lower is better.

23.3% lower daily MAE than the weekly benchmark

LSTM had the lowest weekly-total MAE: 17.343 units.

The stock-planning terms

Lead time	Days until the delivery arrives
Review interval	Extra days covered before the next review
Buffer stock	Extra units kept as a safety allowance
Inventory position	Available stock plus outstanding units

Coverage period = lead time + review interval

Worked example: an order of 38 units

Inputs: stock 50, outstanding 0, lead time 2, review 3, buffer 10

Coverage: 2 + 3 = 5 days

Target stock: forecast 77.52 + buffer 10 = 87.52 units

Amount needed: target 87.52 − inventory 50 = 37.52 units

Order suggestion: round 37.52 up to 38 whole units

How the text assistant understands a question

Question: “What will Product 1 sell next week?”

Text features: TF-IDF turns words into numerical inputs.

Intent classification: logistic regression recognises “forecast”.

Answer: retrieve Product 1’s saved forecast values.

An intent is the type of request the user is making

Questions the assistant can answer

“Show the forecast for Product 2.”

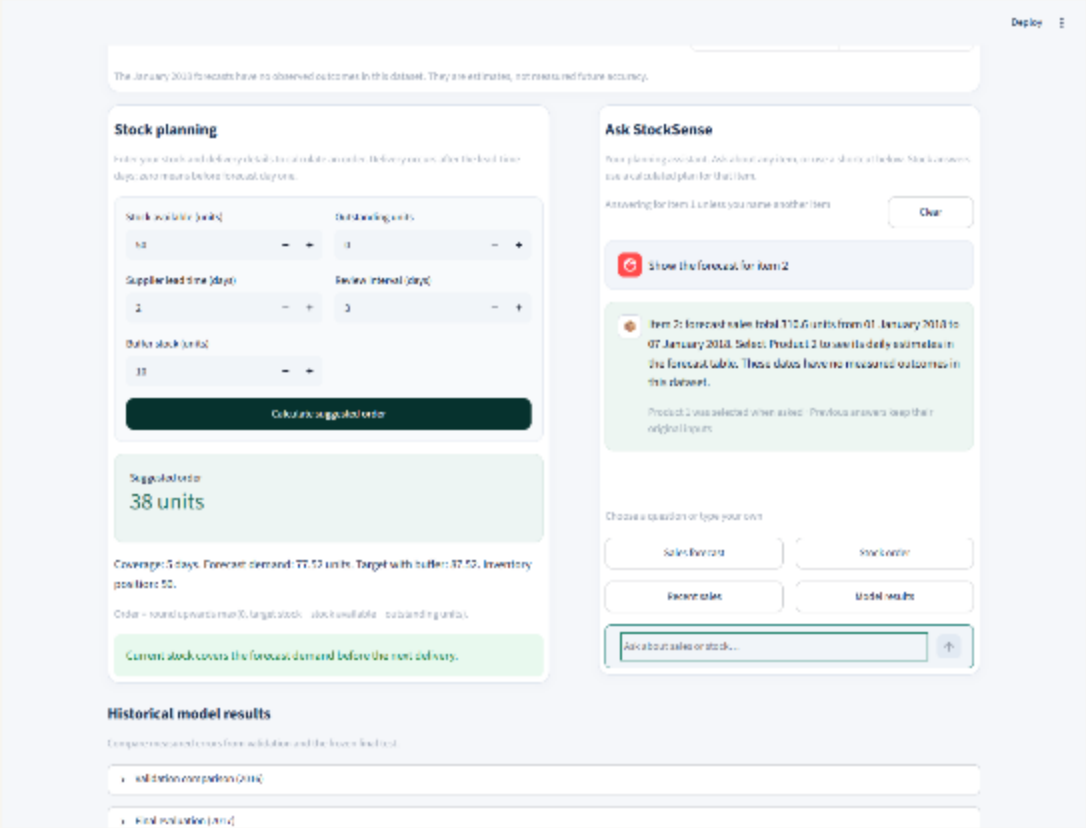
“Compare all five products.”

“Why should I order this amount?”

“What does MAE mean?”

Web and terminal chat use the same forecasts and stock rule

Live application demonstration



1. Select Product 1
2. Show seven dates
3. Calculate 38 units
4. Ask StockSense

Historical demonstration forecast: 1–7 January 2018

Evidence that the prototype works

79

automated tests passed

24/25

supported requests recognised

8/8

unrelated requests rejected

Automated checks verify behaviour. Human usability is still unmeasured.

What the current evidence cannot show

The data is historical and the product names are unknown.

Stock levels and delivery values come from user inputs.

January 2018 forecast outcomes are unavailable.

Real savings and stockout reductions have not been measured.

A future retailer trial would measure stockouts, costs and usability

Conclusion and questions

Random forest achieved 23.3% lower daily MAE than the benchmark.

The stock suggestion shows how each input affects the order.

The web app and terminal assistant make the results accessible.

Questions